

Instead of using a central entity to manage the chain, blockchains use a peer-to-peer network and everyone is allowed to join. When someone joins this network, he gets a full copy of the blockchain. The node can use this to verify that everything is still in order.

When someone creates a new block, it is sent to everyone on the network. Each node would then verify the block to ensure the block has not been tampered with and if everything is fine, each node will add this block to their own blockchain.

All the nodes in this network create consensus. They agree on which blocks are valid and which aren't and reject the blocks that have been tampered with. So, in order to successfully tamper with the blockchain, you would need to tamper with all blocks on the chain, redo the proof of work for each block and take control of more than 50% of the peer-to-peer network, only then will your tampered block become accepted by everyone else, which is practically impossible to do.

This is just a basic idea of how it works and will be further explained in more detail in upcoming chapters.

Applications made on blockchain today are being explored and used in many industries as a cost-effective and secure way to create and manage a distributed database and also to maintain records for all types of digital transactions. 